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EXPERT OPINION

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Applying machine learning techniques for ADME-Tox prediction: a review

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Introduction: Pharmacokinetics involves the study of absorption, distribution, metabolism, excretion and toxicity of xenobiotics (ADME-Tox). In this sense, the ADME-Tox profile of a bioactive compound can impact its efficacy and safety. Moreover, efficacy and safety were considered some of the major causes of clinical failures in the development of new chemical entities. In this context, machine learning (ML) techniques have been often used in ADME-Tox studies due to the existence of compounds with known pharmacokinetic properties available for generating predictive models.

Areas covered: This review examines the growth in the use of some ML techniques in ADME-Tox studies, in particular supervised and unsupervised techniques. Also, some critical points (e.g., size of the data set and type of output variable) must be considered during the generation of models that relate ADME-Tox properties and biological activity.

Expert opinion: ML techniques have been successfully employed in pharmacokinetic studies, helping the complex process of designing new drug candidates from the use of reliable ML models. An application of this procedure would be the prediction of ADME-Tox properties from studies of quantitative structure–activity relationships or the discovery of new compounds from a virtual screening using filters based on results obtained from ML techniques.

Keywords: absorption, distribution, metabolism, excretion and toxicity of xenobiotics, *in silico* drug design, machine learning, molecular modeling, pharmacokinetics, quantitative structure–activity relationships

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1. Introduction

Pharmacokinetics (PK) involves the study of absorption, distribution, metabolism, excretion and toxicity (ADME-Tox) of xenobiotics. It can also be considered the ‘liberation’ term referring to the liberation of a drug from a delivery system, such as a prodrug, polymers or liposomes. The ADME-Tox processes are considered multifactorial and complex because they involve different interactions (i.e., ionic, polar and hydrophobic interactions) with several structures of the human organism, such as membranes, proteins and intra/extracellular environments. As example, simplifying the absorption process as ‘intestinal absorption’, this process depends on several factors such as solubility of drugs, permeability of drugs into the cell membrane, the integrity of the intestinal mucosa, the quantity and quality of ingested food, and so on [1]. The distribution of a drug, for example, depends on the interaction of the drug with plasma proteins, the solubility of the molecule, the portioning between tissues, and so on. Therefore, the generation of mathematical models to represent an ADME-Tox process is a very complex problem.

Article highlights.

- Inappropriate absorption, distribution, metabolism, excretion and toxicity of xenobiotics (ADME-Tox) profile is considered one of major causes of clinical trials failures of new chemical entity and machine learning (ML) techniques have been employed in order to avoid it.
- Both quantitative and qualitative models could be constructed with ML techniques.
- The imminent advances in the research fields of biology, molecular modeling, ML and computer science could strongly impact and provide several unexpected findings in ADME-Tox modeling.
- The quality of a quantitative structure-activity relationships model depends on the ML technique employed, size of data set, variable selection, internal and external validations, as well as the experimental ADME-Tox measurements.
- Support Vector Machines and Partial Least Squares are the most employed ML techniques in ADME-Tox properties.

This box summarizes key points contained in the article.

Aiming to design a new chemical entity (NCE) as a drug candidate, the pharmaceutical industry and the academic area have tried to develop a new compound with good balance between pharmacodynamic and PK profiles, avoiding high rates of failures in drug development. Several authors [2-4] have considered ADME-Tox as one of major causes of drug candidate failure in the drug development process and a good strategy to avoid that is the early detection of bad PK and toxicological profiles. For over a century, medicinal chemists have tried to relate physicochemical properties (solubility, lipophilicity, molecular weight and others) with ADME-Tox characteristics of chemicals [5-8]. Since 1962, quantitative structure-activity relationship (QSAR) models have been developed with the aim of relating chemical data and biological properties and proposing mechanisms of interaction between ligands and biological targets (proteins, nucleic acids and/or membranes), as well as improving the biological activity of diverse series of compounds [9-11]. However, with the advances of powerful computational techniques in the past decades, the QSAR models can also be applied to solve more complex problems, as ADME-Tox profiles of drugs [3]. In this context, both computational chemistry and machine learning (ML) techniques have been successfully used to describe the chemical spaces with thousands of multi-dimensional descriptors and to construct predictive models from the available data sets with significant accuracy.

Several ML techniques have recently broadened the arsenal of tools that can be applied in ADME-Tox studies. Furthermore, some techniques are useful for the establishment of relationships between molecular descriptors and PK properties such as Partial Least Squares, Multiple Linear Regression, Decision Trees, Random Forest, k-Nearest Neighbors, Multilayer

Perceptron neural network, Support Vector Machines, Self-organizing Maps and Principal Component Analysis [12,13], as it is shown in Figure 1. These techniques can be classified as supervised and unsupervised learning and each one has its importance in ADME-Tox studies, as it will be later shown in this review.

One important reason for the increasing number of ML applications in drug discovery processes is that these methods can be appropriately used to construct computer programs that automatically improve with experience in accomplishing a certain task, like classifying a chemical compound as active or inactive to a biological target using its descriptors [14]. Such techniques are also especially useful to treat problems where a data set contains valuable implicit patterns that can be discovered automatically in poorly understood domains where humans might not have the knowledge needed to develop effective algorithms and in domains where the program must dynamically adapt to changing conditions. Furthermore, the possibility of representing chemical compounds with known PK properties as vectors containing information on molecular descriptors makes ML methods well-suited to be applied in ADME-Tox studies. In this review, some studies that apply these techniques in ADME-Tox prediction are presented.

More specifically, from PK properties, it is possible to apply ML techniques for performing classification or regression modeling to help the discovery of new drug-like compounds. In this sense, one can integrate experimental and computational chemistry tools (e.g., calculation of molecular descriptors and construction of pharmacophoric models in combination with ML techniques) for constructing models that receive ADME-Tox properties as input data in order to predict the PK profile of new compounds with desired PK and toxicological profiles (Figure 2). In this way, the main advantage of modeling ADME-Tox properties is the reduction of the number of compounds with undesirable PK and toxicity profile in the early stages of drug design protocols and, therefore, the reduction of failures in further clinical trials [1-3].

2. Trends in ADME-Tox predictions

In 1970s, the pharmaceutical companies spent < 5 billion dollars (USD) with R&D and the number of NCE approved by FDA, per year, was about 10. Since then, the numbers of NCE and costs in R&D have been growing together. In the end of 1990s, the investments in drug design remained increasing but the number of NCE started to decrease after reaching > 50 NCE approved by FDA in 1996. In 2000s, the number of NCE was ~ 25 and the R&D costs reached 30 billion USD [15]. In fact, the R&D investment has been increasing disproportionately to the number of NCE approved by FDA. Decramer and Decramer [16] have affirmed that the pharmaceutical market reached 875 billion of USD in 2010 and costs approximately around 96 billion

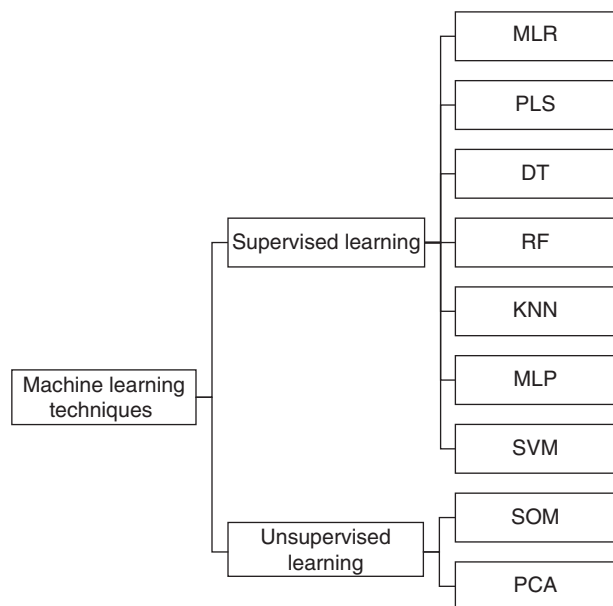


Figure 1. Categorization of ML techniques as supervised or unsupervised learning approaches. As supervised method are highlighted MLR, PLS, DT, RF, KNN, MLP and SVM. Unsupervised methods are SOM and PCA.

DT: Decision trees; KNN: K-Nearest neighbors; MLP: Multilayer Perceptron; MLR: Multiple linear regression; PCA: Principal component analysis; PLS: Partial Least Squares; RF: Random forest; SOM: Self-organizing Maps; SVM: Support Vector Machines.

USD in R&D of drugs. Therefore, there is a clear need for improvement of the ratio between investments and approved NCE.

In the last decades, the pharmaceutical companies have employed some rules based on Lipinski's Rule of Five, Drug-Like, Lead-Like and in-house filters in order to avoid undesirable ADME-Tox and formulation profiles [17,18]. In addition to this fact, there are estimates that, between 2020 and 2030, the computational studies will precede the experimental ones [19], as well as several authors consider the computational approaches in drug discovery as important tools to minimize the cost and mainly risks of failure in the drug development [20]. In fact, the clinical failures have been considered one of several components of a deep valley between drug discovery and NCE approval. However, in the last decades, the PK and toxicological profiles of chemicals submitted to clinical trials have improved. The clinical failures of drug candidates decrease from 40 (1991) to 11% (2000) due to the ADME incompatibility. Unfortunately, the clinical failures caused by toxicological incompatibility increased by twofold (from 11 – 22%), but the main cause for clinical failures is the lack of efficacy (~ 30%) in the same period of time [21]. From 2007 to 2010, clinical failures in Phase II and III are also the main reasons related to the lack of efficacy (51 and 66%, respectively), followed by the lack of safety (~ 20%) and ADME properties, which are responsible

for < 1% [21], indicating that this could be related to the successful application of computational tools (in particular, ML techniques) to predict the PK profile of chemicals in early development states.

In this review, an analysis on the number of published studies in Web of Science, Web of Knowledge and Scopus databases employing ADME-Tox analyses with ML, from 2001 to 2013, demonstrated an increase of 2.7-fold (Figure 3). This data indicates that the improvement in the ADME-Tox profile of the compounds submitted to clinical trials could be related to the appropriate use of ML techniques to analyze PK and toxicological properties in order to prevent clinical failures.

It could be noted that Support Vector Machines is considered by several authors as the most promising technique to be employed in ADME-Tox predictions, as well as in general drug design predictions. Some studies comparing the predictability of ADME properties by using ML techniques can highly support this statement [22-24]. Figure 4 shows the number of the most applied ML techniques, per year, to estimate ADME properties.

In the last 5 years, several tools have been available to perform *in silico* predictions of ADME-Tox properties, for instance, the database of pharmacokinetic data (PK/DB) published and released by Moda *et al.* [25]. This PK/DB contains 1203 compounds with ~ 3000 PK measurements and several predictive PK models. The available predictive models have been constructed with Hologram QSAR (HQSAR) method, and there are models for blood-brain barrier (BBB) permeation [26], human intestinal absorption [27], plasma protein binding (PPB) [28], human oral bioavailability [29], P-glycoprotein inhibition and water solubility [30], previously published by Moda's research group. The available HQSAR models are constructed applying Partial Least Squares method and are internally and externally validated with large data sets (hundreds of compounds). Other relevant online tools to predict ADME-Tox properties can also be mentioned, such as ADMETSAR [31] (prediction of several PK and toxicity end points with classification and regression methods), MetaPrint 2D [32,33] (prediction of metabolic sites), DemQSAR [34] (prediction of volume of distribution and human clearance) and OSIRIS [35] (prediction of mutagenic, tumorigenic, irritant and reproductive affective risks, solubility and druglikeness).

3. ML techniques

ML techniques are concerned with inferring hypothesis from a given training data in order to construct a model that can learn and generalize from previously seen examples. In the context of applying ML techniques to ADME-Tox studies, the focus is on building predictive models that extract patterns from the training data aiming to predict the PK properties of new compounds. Furthermore, several ML techniques are useful for establishing the relationships between molecular descriptors and PK properties, such as Partial Least Squares,

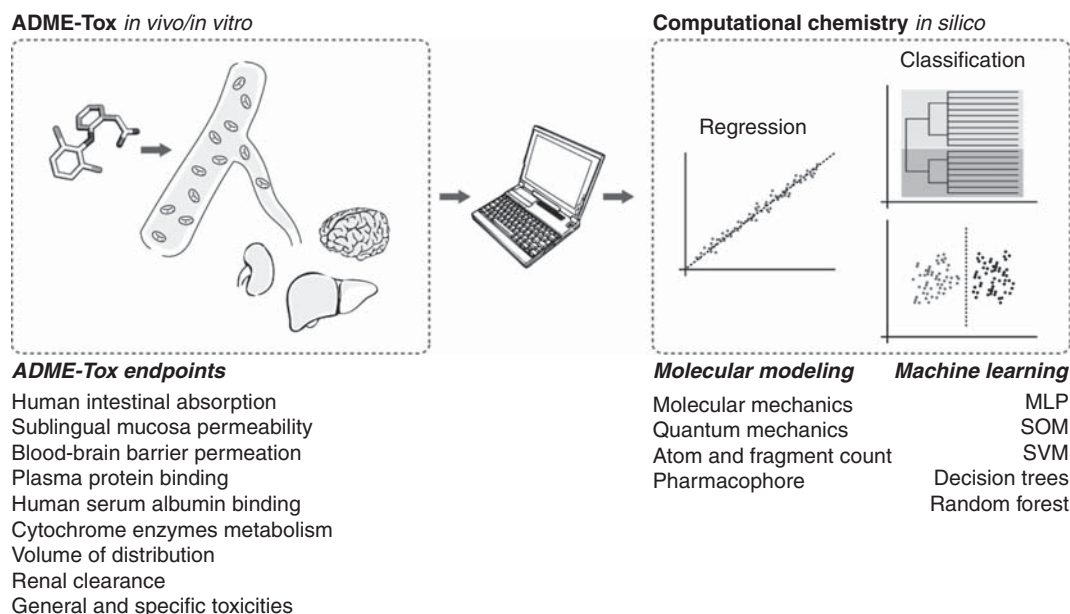


Figure 2. Integration of experimental data and machine learning techniques for modeling ADME-Tox properties.

MLR: Multiple linear regression; SOM: Self-organizing Maps, SVM: Support Vector Machines.

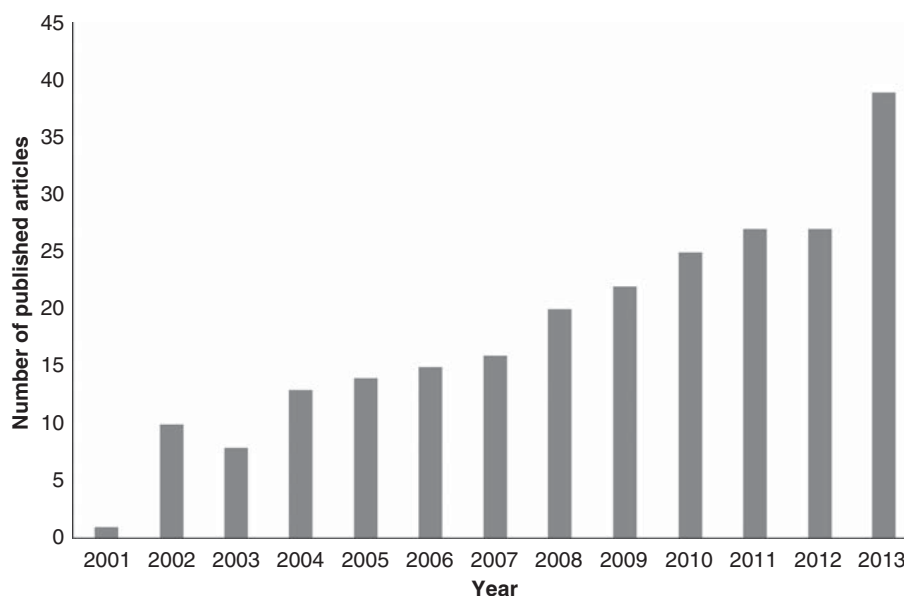


Figure 3. Number of articles employing ML techniques to analyze ADME-Tox properties from 2001 – 2013.

Multiple Linear Regression and Decision Trees [12,13]. In this review, some foundations of the ML techniques most commonly used in ADME-Tox studies are presented. These learning methods can be categorized as supervised or unsupervised methods, as it is shown in Figure 1.

In supervised learning experiments, the task is to create a model from training data that can be used to predict a label

or value (output) that characterizes a new sample (compound) based on the values of its input attributes (chemical descriptors) [36,37]. The term supervised is related to the presence of a ‘teacher’ that provides the desired output (label) for each example. Thus, the external supervisor can assess the ability of the constructed model to predict the output value for new examples. The label (class or category) to be predicted

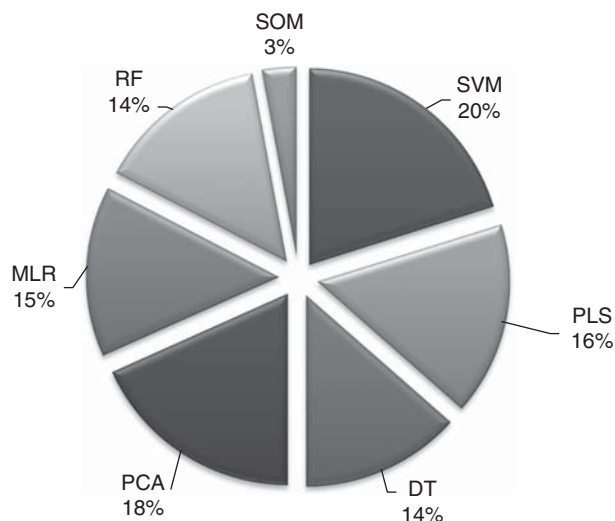


Figure 4. Published articles employing ML techniques to predict ADME-Tox parameters from 2007 – 2013.

DT: Decision trees; MLR: Multiple linear regression; PCA: Principal component analysis; PLS: Partial Least Squares; RF: Random forest; SOM: Self-organizing maps; SVM: Support Vector Machines.

may vary according to the nature of examples (samples or compounds). This label can describe a quantitative (e.g., value/concentration of oral bioavailability) or qualitative (e.g., a classification such as poor-absorption and good-absorption) property that can be learned by regression or classification methods, respectively. In unsupervised learning, there is no external teacher, that is, there are no examples of the function to be learned. Instead, the model should be optimized with respect to some measure that reflects the quality of the obtained solution.

Multiple Linear Regression method has been proposed to model the relationship between independent variables and a dependent variable by fitting a linear equation to the observed data [36]. This technique is simple to implement and is well-suitable to be applied in analyses of small data sets, but it can become problematic if many chemical descriptors in the data set are correlated [38]. In this case, the Partial Least Squares technique can achieve better results by projecting the input variables into a new space [36]. In its general form, this technique creates principal component vectors by maximizing the covariance between different sets of variables and then using these vectors as the basis for the new subspace. Indeed, Partial Least Squares can be regarded as a major generalization of Multiple Linear Regression, since for identical problem formulations, both methods produce identical answers [39]. However, the first technique is particularly appropriate when the matrix of predictors has more variables than observations [38].

K-Nearest Neighbors is a simple ML method based on distance (or similarity) metrics, whereby an unclassified observation (compound) is labeled according to the majority class of

its k -closest instances in the training set [37,40]. In this context, the nearness property is measured by an appropriate distance metric, typically the Euclidean distance. In the first step of the training process, the distances between an unclassified object and all the objects in the training set are calculated and the k most similar ones are selected. Afterwards, the classification process proceeds by labeling the unclassified object with the majority class label of its k nearest training observations. In order to choose the best value for k , it is usual to carry out a cross-validation procedure, such as leave-one-out. A drawback of k -Nearest Neighbors technique is that it does not learn a compact model from the available data, since the training data set has to be stored, so that the classification of a new compound can be carried out. In this sense, this method is computationally expensive when the data set is large. Some advantages of this algorithm are its simple implementation and intuitive operation. Besides, it can also be used in regression problems in a straightforward manner, without significant changes.

Decision Trees is a method that can be used for approximating discrete-valued functions, generating a classification model that is graphically represented by a tree structure [14], as shown in Figure 5A, or equivalently, by a set of if-then rules which can improve the human readability. This learning method is among the most popular ML techniques and has been successfully applied to a broad range of applications [40,41]. Since a model has been created, new observations can be classified by posing a number of questions about their attributes. Each of these questions is stored in an internal node, and every internal node leads to one child node for each possible answer to this question. In this case, each branch descending from that node corresponds to one of the possible values for the attribute being evaluated. In a general formulation, an observation is classified by following the path from the topmost node, named as root, to a terminal node, named as leaf, according to the answers determined by the attribute values of the observation under consideration. The leaf reached at the end of the path provides the classification result. More details about this technique can be found in other studies in literature [14]. Some variations on the originally proposed Decision Trees can be also applied to regression problems.

Although the Decision Trees can produce accurate and interpretable models, it presents some limitations concerning its possible inability to generalize to observations that have not been seen in the training set. Aiming at solving such drawbacks, a method known as Random Forests has been developed. This ML technique combines many Decision Trees to make an ensemble that can be used for classification or regression (Figure 5B) [42,43]. The generalization error of Random Forests converges to a minimum limit as the number of trees in the forest becomes large, and it also depends on the performance of the individual trees in the forest and on the correlation among them.

Multilayer Perceptron neural network is considered the most common neural model used in supervised ML studies

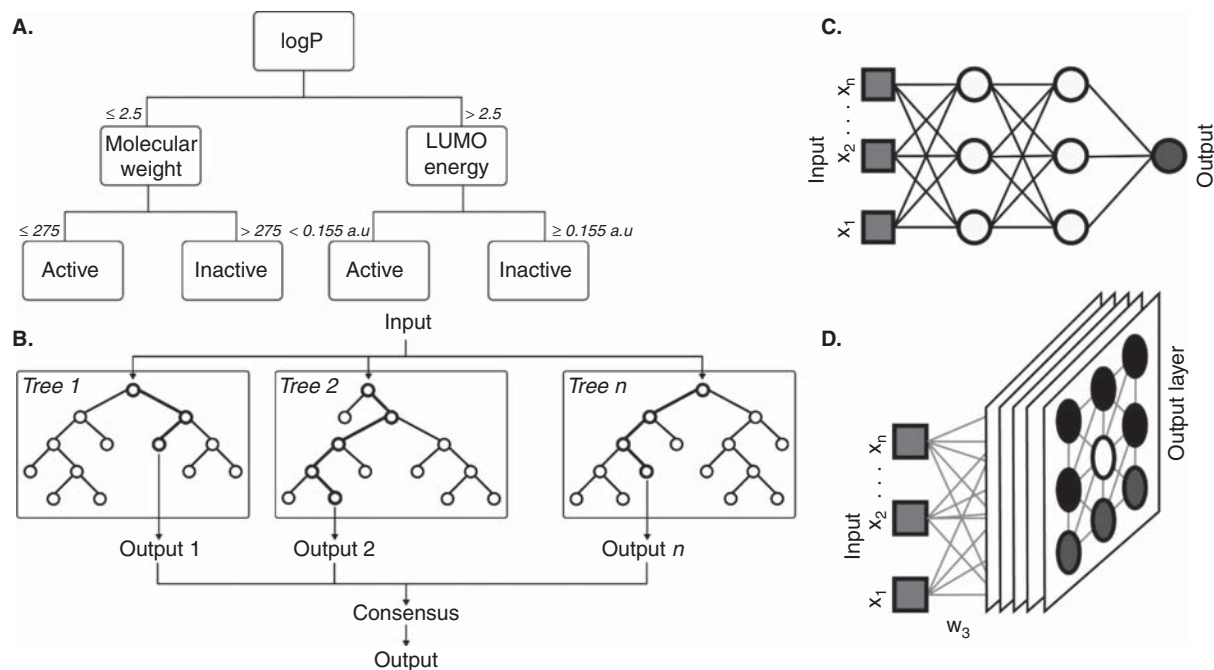


Figure 5. Graphical representation of decision trees (A), random forests (B), Multilayer Perceptron (C) and self-organizing maps (D). LUMO: Lowest unoccupied molecular orbital.

for classification and regression [44]. It has been proved that this neural network is a universal approximator, because it can fit any continuous multivariate function with arbitrary accuracy [45]. As shown in Figure 5C, the Multilayer Perceptron architecture is composed of multiple processing units (called neurons) arranged in consecutive fully interconnected layers, in a way that the first layer consists of the input variables and the last one consists of the output variables, which represent the estimated class (in classification tasks) or the predicted value (in regression tasks). Intermediate layers, also called hidden layers, receive the inputs modified by the passage through the weighted connections. In this model, the training process can be seen as a search for an optimized set of weights for connections that minimize the squared error between the network's outputs and the desired outputs given in the training data set [46].

Support Vector Machines are another supervised ML approach and is based on a statistical learning theory, called Vapnik Chervonenkis theory [47,48]. Actually, this theory aims at proposing computational learning techniques that maximize the generalization ability of classifiers [48]. Roughly speaking, Support Vector Machines consist in building a classification model formally defined by a separating hyperplane that gives the largest minimum distance to the training data. In other words, the goal of this method is to maximize the distance between the separating hyperplane and the nearest sample of each class. Such closest data points are called support vectors and the region of hyperplane bounded by them is known as margin of separation. More details on concepts of

Support Vector Machines can be found in other studies in literature [47,48].

In the unsupervised learning approaches, the problem is to detect patterns from unlabeled data. In this context, the learning process occurs without a 'teacher', because no examples of the function to be learned are given and there is no error measure or reward signal to evaluate a potential solution [49]. Some unsupervised methods widely used in ADME-Tox studies are Principal Component Analysis and Self-organizing Map neural network. Principal Component Analysis is one of the oldest techniques employed in multivariate data analyses [50]. The central idea of this method is to reduce the dimensionality of the original data set while retaining the maximum variation contained in this set. The reduction is obtained by applying a linear transformation to the data in order to derive a set of new variables called principal components, which are uncorrelated and ordered so that the first principal component retains as much as possible the variability of the original data [49,51]. According to Abdi and Williams [51], the four main characteristics of the Principal Component Analysis are: i) to extract the most important information in the data set, ii) to compress the original data set, only storing important information, iii) to simplify the description of a data set, and iv) to allow the structural analysis of observations and variables.

A Self-organizing Map is an unsupervised neural network model that can be used for clustering, visualization and extrapolation. This neural model can be used to study data of high-dimensional spaces by projecting them into a

two-dimensional grid whose main characteristic consists in keeping the topology of the training data set [49]. Typically, the network architecture is composed by an input layer and an output layer where the final result can be observed in a grid of connected neurons, as shown in Figure 5D. Here, the learning scheme is based on a competitive strategy, known as ‘winner takes all’, by which if a particular neuron is chosen to represent a given input, then its neighbors are also stimulated in order to represent similar patterns. The result of this process corresponds to an input data projection, so that points that are adjacent in the high-dimensional space will also be adjacent in the lower-dimensional output grid, called Kohonen map [46].

4. Applications of ML techniques in ADME-Tox studies

Since 2007, several studies have been conducted by employing ML techniques to predict ADME-Tox properties. In general, besides the prediction of PK and toxicological parameters, these researches can also provide several evidences of mechanisms for each individual process. Therefore, *in silico* researches in ADME-Tox field have provided valuable results and insights that can be applied to prevent clinical failures and that can help this science to advance.

4.1 Absorption modeling

The modeling of absorption depends on the administration route of the investigated drug class. For example, Caco-2 (a human colonic adenocarcinoma cell line) and MDCK cells (Madin Derby Canine Kidney) have been used to predict *in vitro* drug transport through the intestinal epithelia when a drug is orally administered [52,53]. Moreover, a drug can be administered by other via, such as sublingual, intranasal, inhalation, ocular, rectal, subcutaneous, intramuscular and intravenous via (this is considered without absorption).

Shinde *et al.* [54] constructed two QSAR models for Caco-2 cell permeability using 60 drug-like compounds and employed membrane interaction QSAR (MI-QSAR) and HQSAR methods aiming to describe the molecular insights on the human intestinal absorption. The MI-QSAR model was constructed employing multidimensional linear regression and genetic function approximation; HQSAR model was constructed employing Partial Least Squares technique. The two constructed QSAR models showed good statistical parameters. From the physicochemical interpretation of the models, the authors described that small compounds and substances with high dipole moment have good permeability; the presence of chlorine atoms, methyl and polar groups (responsible for hydrogen-bonding) can negatively influence the membrane permeability. The MI-QSAR method was also employed to predict skin penetration [55], corneal permeability [56], human oral absorption [57], gastrointestinal cell permeation of organic compounds [58] and BBB partitioning [59].

Goswami *et al.* [60] constructed an *in silico* model to estimate the drug permeability across sublingual mucosa (LogP_s) employing calculated and experimental descriptors. The authors measured the LogP_e values at the same experimental conditions and constructed a QSAR model employing the number of hydrogen-bond donor groups and LogD descriptors using Multiple Linear Regression. The constructed model showed good predictive ability and an excellent fit between experimental and predicted permeability. However, the results were considered statistically poor because of the low number of observations (14 known drugs – Lidocaine, Propranolol, Atenolol, Verapamil, Diltiazem, Amitriptyline, Bupivacaine, Pindolol, Nimesulide, Furosemide, Naproxen, Warfarin, Caffeine and Antipyrine) and due to the absence of some important statistical tests. Despite of the weak-points of the constructed model, this study is pioneer in the construction of an *in silico* sublingual mucosa permeability model and it is expected that it encourages future investigations to improve the prediction of this absorption parameter for the development of new drugs.

Self-organizing Maps can generate topological feature maps that are useful to analyze the shape and surface properties of the objects (chemical substances) responsible for the PK properties. In an oral bioavailability study of 805 structurally diverse drug and drug-like molecules, Xu *et al.* [61] applied Self-organizing Maps to divide a set of compounds in training and test sets, ensuring greater consistency between them.

Despite of all the studies in this area [54-60], Antunes *et al.* [62] consider that the lack of reliable information for permeability and intestinal absorption is the major impediment for the successful of absorption modeling.

4.2 Distribution modeling

The plasma is an aqueous solution with ~ 7% of proteins and corresponds to 55% of human blood. The main proteins of human plasma are albumin, α -1-acid glycoprotein, globulins, clotting factors, regulatory proteins and lipoproteins (e.g., high-density lipoprotein, low-density lipoprotein and very low density lipoprotein). Different drugs reversibly bind to different plasma proteins and this affects the volume of distribution, elimination and the pharmacologic effect since only the unbound drug molecules are able to pass the membranes and achieve its biological target [63,64]. Since 2000s, QSAR models have been constructed to predict human PPB [22,65], but in the last 5 years Ma *et al.* [66] and Ghafourian and Amin [67] developed models based on Support Vector Machines and boosted tree methods, which showed a good predictive power.

Chen and Chen [68] constructed models to predict human serum albumin (HSA) binding employing physicochemical and docking descriptors, as well as Multiple Linear Regression and Support Vector Machines. The HSA binding is an important parameter to the distribution modeling, because HSA is a plasma protein and when compounds bind to this protein its concentration to reach the biological target decreases. The model based on Support Vector Machines

had better results than Multiple Linear Regression to construct QSAR models with both physicochemical and docking-derivative descriptors.

4.3 Metabolism modeling

The metabolism is an important step of the PK process because it is possible to predict the formation of metabolites and, consequently, to predict if the metabolites are bioactive, toxic or quickly eliminated. The drug metabolism is very complicated because it involves several CYPs at the Phase I metabolism and several other enzymes, such as UDP-glucuronosyltransferases, sulfo and methyltransferases at the Phase II metabolism. In general, as the metabolism reactions are mediated by enzymes, there are several research groups employing molecular docking, molecular dynamics and QM/MM techniques to predict the metabolism of chemicals. Several reviews have discussed some aspects of structure-based ADME-Tox predictions [69,70]. From these techniques, it is possible to study both molecular and electronic mechanisms involved in these reactions, while, from ML methodologies, it is possible to predict and classify if a compound will or will not be metabolized by determined enzymes.

Athersuch *et al.* [71] constructed models to classify and predict the metabolism of substituted anilines in order to study individual pathways of metabolism. The authors employed Principal Component Analysis, Soft Independent Modeling of Class Analogy and Partial Least Squares discriminant analysis to create the models to classify 46 compounds: 23 anilines and their *N*-acetylated derivatives. The best model obtained to discriminate *N*-acetylated and *N*-oxanilic acids, employed the compound set and various semi-empirical physicochemical properties (calculated with MOPAC). Then, the authors tested the effect of solvent calculating the physicochemical parameters with *ab initio* method (HF/3.-21G) and continuum solvation model (PCM) with Gaussian03 software. The model generated with the simulation of *in vivo* environment showed a good result for the same data set. Therefore, the authors concluded that an accurate method of solvation can improve the QSAR models to predict a specific metabolism.

4.4 Excretion modeling

The main route of human xenobiotic excretion is from kidneys (urine), but there are other routes of excretion, such as feces and bile, sweat and breath. The excretion pathways directly impact the amount of available drug molecules to interact with the biological target, as well as the half-life time and the administered dose.

Paine *et al.* [72] applied Partial Least Squares and Random Forest to predict the human renal clearance for 349 compounds. In the studied data set, 59 compounds are acids, 124 are bases, 112 are zwitterions and 54 are neutral compounds. In all data set, the Random Forest model showed superiority in all statistical parameters (fitness, robustness and predictability). Hsiao *et al.* [73] also compared the

regression models obtained from Random Forest with other ML methodologies, such as Orthogonal Partial Least Squares and Multiple Linear Regression, to predict clearance of 244 drugs and the Random Forest model showed better results than the two other methods.

4.5 Toxicity modeling

The toxicity is a complex biological process because it depends on several different end points. The toxicity processes can occur at organelle, cellular and systemic levels and can result from receptor/enzyme or DNA interaction, induction or inhibition of hepatic metabolism, as well as the combination of several causes.

Zhang *et al.* [74] constructed a classification model to predict mitochondrial toxicity using Support Vector Machines. In this study, the authors employed Genetic Algorithms for the feature selection and conjugate gradient method to optimize the model parameters. Employing a data set with 288 drugs or drug-like compounds, the best constructed model achieved optimal values of accuracy for training and test sets. The mitochondrial toxicity is related to several side effects of drugs, as well as long-term toxicity as lipodystrophy, lactic acidosis, neurological and cardiovascular disorders.

Different from the previous study, which focused on the toxicity of an organelle, Myshkin *et al.* [75] constructed several classification models to predict the organ toxicity of compounds. Several properties as general hepatotoxicity and nephrotoxicity, as well as specific liver and kidney necrosis, liver and kidney relative weight gain, liver lipid accumulation and nephron injury were assessed by these models. The models were generated using Decision Trees method and showed good performance. There are several reviews to better understand the application and evolution of QSAR analyses used to evaluate toxicological effects in the last decades [76,77]

4.6 QSPR studies of other properties related to ADME-Tox

Water solubility is an important parameter in the drug development because is related to some properties of a compound, such as formulation, intestinal absorption, distribution and elimination. Toropov *et al.* [78]. employed SMILES-based (Simplified Molecular Input-Line Entry System) descriptors to construct a QSPR model regarding water solubility. The 1311 compounds in the data set were distributed in five random subsets; these sets were selected according to the chemical similarity and the QSPR models were constructed employing CORAL software and Ordinary Least Squares analysis. In comparison with other QSPR models for water solubility, the model constructed by Toropov *et al.* had an excellent improvement in the internal robustness and external predictive power. Table 1 describes several QSPR models for water solubility employing large data sets (usually, > 1000 compounds), which can be considered statistically reliable based on the values of q^2 , r^2 and r^2_{test} .

Table 1. Comparison of several QSPR models regarding water solubility for large compound sets.

Author	Year	N	q^2	r^2	r^2_{test}
Toropov <i>et al.</i> [78]	2013	1311	0.92	0.96	0.94
Delaney [81]	2004	2874	-	0.72	-
Hou <i>et al.</i> [82]	2004	1290	0.90	0.92	0.88
Yan and Gasteiger [83]	2003	1314	0.92	0.94	0.85
Butina and Gola [84]	2003	3328	-	0.80	0.74
Gao <i>et al.</i> [85]	2002	1139	0.91	0.92	0.91
Tetko <i>et al.</i> [86]	2001	1312	0.92	0.95	0.90
Huuskonen [87]	2000	1318	0.92	0.94	0.91

From Table 1, it is possible to observe that the predictive power of the QSAR models for the prediction of water solubility has increased along the 13 last years. In summary, the increase of the statistical quality of the models for water solubility affects directly the accuracy of the prediction of this property. Therefore, the newest models could be employed to predict the water solubility of unknown compounds in order to help the improvement of the PK profile of drug candidates.

Zhang *et al.* [79] also constructed models for solubility and lipophilicity, employing 1342 and 10781 compounds, respectively. In this study, the authors classified the compounds in three classes according to the logS and logP values, as shown in Table 1. Support Vector Machines combined with Genetic Algorithms (feature selection) and Conjugate Gradient method (used for parameter optimization) was employed to construct the models. The generated model for logS was compared to K-Nearest Neighbors and consensus neural network-based methods and the prediction accuracy for the training set was equal to 94.2% (for the Support Vector Machines-based models), 80% (for the K-Nearest Neighbors-based models) and 87.2% (Consensus neural network-based models). The accuracy of the logP model for the training set was equal to 89.8%. The obtained results indicate that the models based on Support Vector Machines were suitable for the solubility and lipophilicity prediction.

Zhang *et al.* [80] constructed a model to classify multi-drug-resistance protein 2 inhibitors and non-inhibitors using Support Vector Machines. The overexpression of this protein is related to the decrease of efficacy of several drugs. Sixteen molecular descriptors were selected using Genetic Algorithm. The Support Vector Machines model was constructed using 257 compounds (training set) and an independent test set with 61 compounds was used to validate the model; it was obtained an accuracy of 82.9 and 77.1% for the training and test sets, respectively.

5. Conclusion

PK (ADME-Tox) processes are considered multifactorial and complex since they involve different interactions with several

structures of the human organism, such as membranes, proteins and intra/extracellular environments. The evolution of computational approaches has resulted in an increase of the number of successful applications of ML techniques in PK studies. The integration of experimental and computational tools has contributed to an effective route in the drug discovery and design. In this study, a brief description and various applications of some ML techniques used in ADME-Tox studies, such as Partial Least Squares, Multiple Linear Regression, Decision Trees, Random Forest, k-Nearest Neighbors, Multilayer Perceptron, Support Vector Machines and Self-organizing Maps have been presented. An analysis on the number of published articles in Web of Science, Web of Knowledge and Scopus databases employing ADME-Tox analyses with ML, from 2002 to 2013, has showed an increase of 2.7-fold. Therefore, from this review, it is possible to conclude that many ML methodologies can be considered as promising techniques to be applied in ADME-Tox predictions, as well as obtaining lead compounds with good PK profile.

6. Expert opinion

An ideal QSAR/SAR model for estimating ADME-Tox parameters must highly be predictive and accurate, as well as constructed with simple molecular descriptors in order to be applied in virtual screenings as filters in the initial steps of pharmaceutical R&D. This scenario should avoid discarding of compounds with a good ADME-Tox profile, as well as including false positives in the screening protocol. If the model is constructed with molecular descriptors derived from an expensive computational approach, large chemical libraries (with over than a million of compounds) cannot be filtered with this methodology. From several studies that have applied ML techniques to ADME-Tox properties, one can closely observe critical points in the model generation that can directly affect the performance of each ML technique. In order to overcome critical points described in this work, special attention should be given to new techniques for variable selection and resampling data, which could be considered in novel applications to the PK studies. In this review, it could be noted the growing use of Support Vector Machines for the prediction of ADME-Tox properties. However, one should not discard other ML techniques for applications in new data sets, because these techniques have also showed promising results in ADME-Tox studies.

Indeed, the molecular modeling of a well-defined biological end point has many difficulties. More specifically, the ADME-Tox profile depends on several complex biological mechanisms, which make its modeling a very delicate and complicated task. However, the recent increase of available experimental data regarding ADME-Tox properties, as well as the development of new refined ML techniques that can be successfully applied for handling critical points in a QSAR modeling have evidenced important trends towards

the generation of accurate ADME-Tox models. Another promising alternative to increase the accuracy and applicability of ADME-Tox models (in special, toxicity models) is through the definition of more specific end points to model. Thus, the property to be modeled should become simpler than an unspecific one (for instance, kidney necrosis and nephrotoxicity) and it could contribute for decreasing the number of embedded errors in this biological data. Furthermore, the emergence of data from uncommon routes of administration and other measurements can provide new fields that can be explored in the future.

One of the main trends in ADME-Tox modeling is to create and apply a unique protocol employing the combination of suitable ML techniques for each specific problem in order to obtain an optimal and robust QSAR model. Other important tendency in this area is the combination of QSAR studies and other molecular modeling techniques, such as QM/MM approach, as mentioned in the metabolism section. Finally, the development of accurate models to predict molecular

and electronic properties (i.e., new quantum approaches) and the increase of computational power can provide the capacity to process and calculate more precise descriptors that could be used as input data to obtain QSAR models. In summary, the imminent advances in the research fields of biology, molecular modeling, ML and computer science could strongly impact and provide several unexpected findings in ADME-Tox modeling.

Declaration of interest

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